

QUIZ-1
Marks. 3*5 =15

1. Write down the steps of data analysis process.
2.
 - i. Write 2/3 examples of first party data (collecting data) and justify.
 - ii. In a survey, respondents were asked for their favorite fruit: Apple, Banana, or Mango. What type of categorical data is this, and why?
3. Find the Variance and Standard deviation of the following numbers: 1, 2, 4, 5, 7, 8, 9, and 11.

Q-2, IIT (7+6+2)

1. The sales of a company (in million dollars) for each year are shown in the table below.

Year (x)	2005	2006	2007	2008	2009
Sales (y)	12	19	29	37	45

- Use the least squares regression line as a model to estimate the sales of the company in 2012.

2. Study the Data set and solve these (Use python language):

- i. Show the first 15 data sets from the given data set.**
- ii. From the column of “Salary”, find the wrong values if values are greater than 1,00,000. Then correct those values by replacing the value 55,000.**
- iii. In the column of “Salaries” find the blank cells and fill those cells using mean value.**

3. Define a data mart. Distinguish between a dependent data mart and an independent data mart with suitable examples.

Q.3, IIT

1. How does the choice of k affect the bias-variance tradeoff in KNN classification, and what risks arise when k is too small or too large?
[3]

2. KNN is considered a “lazy learner or eager learner.” What does this mean in practice, and what are the advantages and disadvantages of this property?
[3]

3. Point: {1, 3, 5, 7, 9, 11} with classes {B, A, B, A, B, A}

For test point $x = 7$, determine the predicted class for $k = 1$, $k = 3$, and $k = 5$.

[5]

4. i. Using an example, state the triple constraint of IT project management.
[4]
- ii. List out the core functions name of IT project management.